Rupankar Dutta

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GitHub LinkedIn

Profile

- B.Tech student at KIIT University interested in robotics, embedded systems, IoT systems, and software engineering.
- Hands-on experience building IoT devices, REST APIs, and machine learning models using Python and C.

Education

Kalinga Institute of Industrial Technology (KIIT) 2023–2027
Bachelor of Technology CGPA: 8.4/10

Skills

- **Programming:** Python, C
- **Backend Development:** FastAPI, Flask, REST API Development
- **Machine Learning / NLP:** Linear Regression, LSTM, TextBlob, Scikit-learn
- **Data Processing:** Web Scraping, Data Analysis, API Development
- **Embedded Systems:** Arduino, ESP32, IoT System Development

Technologies

- **Databases:** MySQL
- **Libraries:** BeautifulSoup, PyPDF2
- **Framework Ecosystem:** Pydantic, Swagger
- **Development Tools:** Git, VS Code
- **Platforms:** TensorFlow, ThingSpeak

Projects

- **Dynamic Programming Text Auto-Correction System (C)** – Built a spelling correction system using dynamic programming techniques to compare user input against valid words and suggest likely corrections. Improved typo detection efficiency through optimized string matching and edit-distance-based comparison logic.
- **Automated Plant Watering System (ESP32, IoT)** – Developed a smart irrigation system using ESP32, soil moisture sensors, and DHT11 to monitor environmental conditions and automate watering. Integrated OLED-based live status display and ThingSpeak cloud logging for real-time observation and remote tracking.
- **WiFi Network Scanner (ESP32)** – Created an ESP32-based wireless scanning tool to detect nearby WiFi networks and measure signal strength values. Displayed SSIDs and RSSI information on an OLED screen, making the device useful for quick network diagnostics and signal analysis.
- **Sentiment Prediction API (FastAPI, TextBlob)** – Designed a REST API for sentiment classification of text into positive, negative, and neutral categories using TextBlob. Added request validation with Pydantic and interactive testing through FastAPI Swagger documentation to improve reliability and usability.
- **Flask Authentication Dashboard (Flask, MySQL)** – Built a secure web application with user registration and login features using Flask and MySQL. Implemented password hashing, session handling, and a responsive dashboard for storing and displaying authenticated user information safely.
- **Conversational Chatbot with PDF Reader (Python)** – Developed a rule-based chatbot capable of extracting and responding to information from PDF documents and web pages. Combined PyPDF2, BeautifulSoup, and keyword matching techniques to support simple document querying and web content retrieval.
- **Stock Price Prediction System (Machine Learning)** – Implemented time-series forecasting models using Linear Regression and LSTM to analyze historical stock data and predict future price trends. Built a Flask-based interface for displaying outputs and visualizing predicted movements through interactive graphs.